

Parathyroid Hormone (PTH), Intact

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

This test is used for diagnosis of parathyroid disease and other diseases of calcium homeostasis and for monitoring patients undergoing renal dialysis.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Custom Additional Information

PTH is an 84-amino-acid peptide hormone which is responsible for the regulation of serum calcium levels within a narrow range. PTH is secreted in response to decrease in serum calcium levels by increasing the renal reabsorption of calcium and lowering reabsorption of phosphorus. The measurement of PTH is a very useful tool in the differential diagnosis and management of hypercalcemia. PTH assays can be of help in the diagnosis of tumors and hyperplasia of the parathyroid gland, as well as in localizing hyperfunctioning parathyroid tissue by assay of samples obtained via venous catheterization. The native or intact (1-84) PTH has a short half-life, measured in minutes, whereas the carboxy and midmolecule fragments, which are biologically inactive, have half-lives 10- to 20-fold higher. The high concentrations of biologically inactive fragments have interfered with use of C-terminal or midmolecule assays for evaluation of parathyroid function in patients with impaired renal function. Intact PTH assays provide a more accurate assessment of parathyroid patients including those with various renal diseases.

Specimen Requirements

Specimen

Plasma (preferred)

Volume

1.5 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Lavender-top (EDTA) tube (preferable)

Collection Instructions

Transfer separated plasma into a plastic transport tube clearly labeled as "EDTA Plasma" and maintain at refrigerated temperature.

Stability Requirements

Temperature	Period
Room temperature	2 days (stability determined by manufacturer or literature reference)
Refrigerated	3 days (stability determined by manufacturer or literature reference)
Frozen	6 months (stability determined by manufacturer or literature reference)
Freeze/thaw cycles	Stable x1 (stability determined by manufacturer or literature reference)

Reference Range

See table.¹

Intact PTH (pg/mL)	Calcium (mg/dL)	Interpretation
15–65	8.6–10.2	Normal
>65	>10.2	Primary hyperparathyroidism
>65	<10.2	Secondary hyperparathyroidism
<65	>10.2	Nonparathyroid hypercalcemia
<15	<8.6	Hypoparathyroidism
15–65	<8.6	Nonparathyroid hypocalcemia

Storage Instructions

Refrigerate (preferable)

Causes for Rejection

Whole blood; plasma other than EDTA; hemolysis; serum at room temperature

References

Brown RC, Aston JP, St John A, et al. Comparison of poly- and monoclonal antibodies as labels in a two site immunochemiluminometric assay for intact parathyroid hormone. *J Immunol Methods*. 1988; 109(1):139-144.

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Endres DB, Villaneuva R, Sharp CF Jr, et al. Immunochemiluminometric and immunoradiometric determinations of intact and total immunoreactive parathyrin: Performance in the differential diagnosis of hypercalcemia and hyperparathyroidism. *Clin Chem*. 1991; 37(2):162-168.

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Kao PC, van Heerden J, Grant CS, et al. Clinical performance of parathyroid hormone immunometric assays. *Mayo Clin Proc.* 1992; 67:637-645. PubMed 1434896 (<http://www.ncbi.nlm.nih.gov/pubmed/1434896>)

Nussbaum SR, Zahradnik RJ, Lavigne JR, et al. Highly sensitive two site immunoradiometric assay of parathyrin, and Its clinical utility in evaluating patients with hypercalcemia. *Clin Chem.* 1987; 33(8):1364-1367. PubMed 3608153 (<http://www.ncbi.nlm.nih.gov/pubmed/3608153>)

Footnotes

1. Soldin SJ, Brugnara C, Gunter KC, et al. *Pediatric Reference Ranges*. 2nd ed. Washington, DC: AACC Press;1997:119.